# PlanRead Sample Construction Glossary

Sample retrieval corpus for the PlanRead demonstration build. Definitions written for this project and released under the MIT licence; this file is not the RSMeans Illustrated Construction Dictionary.

## Reading a plan set

A construction plan set is an ordered collection of sheets that together describe one building. The cover sheet carries the project title, the list of sheets, the design loads and the general notes. Architectural sheets follow, usually numbered A-0 upward, then structural, mechanical, electrical and plumbing disciplines each with their own prefix.

Floor plans are drawn as a horizontal cut through the building at roughly four feet above the floor, so that doors, windows and partitions all appear in the same view. Walls are cut and therefore drawn heavy, with a solid or hatched fill between two parallel lines. Anything above the cut plane, such as roof framing or the outline of upper cabinets, is drawn with a lighter dashed or thin line.

Elevations show the outside faces of the building; sections show a vertical cut, usually keyed to a section mark on the plan. Detail callouts enlarge a small part of a plan or section at a larger scale, and the callout bubble tells the reader which sheet and which detail number to turn to.

## Common line and annotation conventions

Dimension strings are drawn as a continuous line with tick marks at each end and the measured value written above the line, normally in feet and inches. Dimensions read from the bottom or the right of the sheet wherever possible, so a dimension set at ninety degrees to normal reading direction is a sign that the view has been rotated.

A keyed note is a short leader line ending in a dot or arrow on the object, with the text at the other end. A schedule is a table that lists repeated objects, such as doors, windows or rooms, by tag; the tag on the plan is a short code that points into the table, so the plan stays uncluttered and the detail lives in one place.

A specification is the written counterpart to the drawings. Where the two disagree, the specification usually governs materials and workmanship, and the drawing governs geometry and location. The general notes on the cover sheet state which document takes precedence for that project.

## Term definitions

LINTEL. A horizontal structural member spanning an opening such as a door or window, carrying the load of the construction above and transferring it to the walls on either side. On a drawing the lintel is usually shown as a heavy line across the opening, often with a keyed note giving its material and size.

SILL. The horizontal bottom member of a window or door frame, or the sloping member at the base of a wall that sheds water away from the building. On a window schedule the sill height is normally given as a dimension from finished floor, because it controls both daylight and fall protection.

HEADER. A horizontal member that collects loads over an opening and distributes them to adjacent framing, structurally similar to a lintel but in wood-frame construction usually built up from multiple plies. A header schedule lists the span, the number of plies and the jack studs required for each opening size.

FOOTING. The widened concrete base at the bottom of a foundation wall or column that spreads the building load over a larger soil area. Footings are dimensioned on the foundation plan with width, depth and reinforcing, and their depth below grade is governed by frost line requirements.

JAMB. The vertical side member of a door or window opening, forming the face that the door leaf closes against. On an enlarged plan or door schedule the jamb detail indicates the frame type, the rough opening and how the frame meets the wall finish.

SOFFIT. The underside of a projecting element, most commonly the underside of an eave, a beam or an overhanging cabinet run. A soffit is usually called out where two planes meet and the joint must be closed against weather or where a finish surface changes.

RISER. The vertical face of a stair step, measured from the top of one tread to the top of the next. Riser height and tread depth together determine stair geometry, and building codes cap the riser height and the variation between consecutive risers.

TREAD. The horizontal walking surface of a stair step, measured from the front edge of one riser to the front edge of the next, excluding any nosing. Stair dimensions on a plan are normally given as a run and a rise for the whole flight, with the individual tread and riser noted once.

REBAR. Reinforcing bar embedded in concrete to carry tensile stress, specified by bar size, spacing and cover. On a foundation or slab plan, reinforcing is called out with a leader to the region it applies to, together with the required clear cover from the concrete face.

FLASHING. A thin impervious material installed to prevent water from penetrating at a junction, such as a roof-to-wall intersection, a parapet or a window head. Flashing details always show a lap and an uphill turn, because the material only works if water is directed back toward the drainage surface.

EAVE. The lower edge of a sloping roof that overhangs the wall, consisting of the fascia, the soffit and the roof edge. On a section the eave detail shows the ventilation path into the attic and how the roofing terminates over the wall.

RIDGE. The highest horizontal line of a sloping roof, where two opposing roof planes meet. The ridge is shown on the roof plan as a heavy line, and the ridge board or beam that supports the rafters at that line is detailed in the framing section.

STUD. A vertical framing member in a wall, typically spaced at sixteen or twenty-four inches on centre. Stud spacing, size and grade are usually stated in a general note or a wall type legend rather than repeated on every wall of the plan.

JOIST. A horizontal framing member that carries floor or ceiling loads between beams or bearing walls, repeated at regular spacing. Floor framing plans show joist direction with an arrow and give the size, spacing and span in a single note.

SHEATHING. The panel material fastened over studs or joists to provide a nailing surface and to brace the frame, such as plywood or oriented strand board. Sheathing thickness and rating appear in the wall section or the structural notes, not on the floor plan.

GRADE BEAM. A concrete beam cast at or near grade that ties foundation walls or pile caps together and carries loads across weak or expansive soil. It is drawn on the foundation plan with its width, depth and reinforcing called out separately from the wall footing.

CONTROL JOINT. A deliberate plane of weakness formed or sawed into a concrete slab or wall to control where cracking occurs. Control joints are shown on the plan as a dashed line with a note giving spacing and depth, because the crack will happen whether or not the joint is drawn.

WEEP HOLE. A small opening left in a mortar joint or a cladding system to let trapped water drain out of a cavity wall or above a flashing. Weep holes are drawn on masonry elevations as a short gap at the base course, usually at regular spacing.

SPECIFICATION. The written project document that sets out materials, products, execution and quality standards for each work item, organised by division. A drawing note that says 'see specification' points to a numbered section there, and the drawing is not complete without it.

SCHEDULE. A table on a drawing sheet that lists repeated elements with their attributes, such as doors, windows, rooms or finishes. The plan carries a short tag for each element and the schedule carries the full description, so a change is made once in the table rather than in every occurrence.

LEGEND. A key on the sheet that explains the symbols, hatch patterns, line types and abbreviations used on that drawing. A legend is sheet-specific in many offices, so the same symbol may mean different things on architectural and structural sheets.

DETAIL CALLOUT. A circular or oval symbol placed on a plan or section that refers to an enlarged drawing elsewhere in the set. The top of the bubble gives the detail number and the bottom gives the sheet number, so the reader can follow the reference without searching the whole set.

SECTION MARK. A symbol, usually a circle bisected by a line with a letter or number at each end, that shows where a building section is cut and which direction the section looks. The matching section drawing carries the same tag so the two can be read together.

PLAN VIEW. An orthographic projection looking down on the building, showing layout in the horizontal plane. A floor plan is a plan view cut at a fixed height, while a roof plan is a plan view from above the roof, and the two are not interchangeable.